

MCU (ESP32-S3): Working Document

Digital Audio Player • Rev 13

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1. Summary

Field	Value
Manufacturer / MPN	Espressif Systems, ESP32-S3 (base chip)
Package	56-QFN, 7 × 7 mm, exposed pad (pin 57, GND)
Memory	No in-package flash; no PSRAM (not used); external 8 MB quad SPI NOR flash, GigaDevice GD25B64EWIGR
Function	Main controller: file decode, I2S to the DAC, UI, charger control, USB mass storage
Clock	External 40 MHz passive crystal (compulsory), ECS-400-10-37B2-CKY-TR, $C_L = 10 \text{ pF}$

Source: [2], [3], [4], [11]

2. Design Requirements

Requirement	Value	Source
Supply	+3V3_DIG; 3.0 to 3.6 V on all supply pins	[2]

Requirement	Value	Source
Crystal	40 MHz only; accuracy within ± 10 ppm	[2]
Storage	microSD over 1-bit SDMMC, card detect	Author
Audio	I2S slave to the ES9038Q2M (DAC is master)	Author
Control	I2C to the DAC and the BQ24250 at 100 kHz	Author
Display	4-wire SPI to the ER-TFT024IPS-3	Author
USB	USB-C, mass storage (GPIO19 / GPIO20)	Author, [2]
Constraints	Low noise · Small footprint	Author

3. Pin Configuration

3.1 IO Pin Default Configuration

No.	Name	Power	At Reset	After Reset
1	LNA_IN			
2	VDD3P3			
3	VDD3P3			
4	CHIP_PU	VDD3P3_RTC		
5	GPIO0	VDD3P3_RTC	IE, WPU	IE, WPU
6	GPIO1	VDD3P3_RTC	IE	IE
7	GPIO2	VDD3P3_RTC	IE	IE
8	GPIO3	VDD3P3_RTC	IE	IE
9	GPIO4	VDD3P3_RTC		
10	GPIO5	VDD3P3_RTC		
11	GPIO6	VDD3P3_RTC		
12	GPIO7	VDD3P3_RTC		
13	GPIO8	VDD3P3_RTC		
14	GPIO9	VDD3P3_RTC		IE
15	GPIO10	VDD3P3_RTC		IE
16	GPIO11	VDD3P3_RTC		IE
17	GPIO12	VDD3P3_RTC		IE
18	GPIO13	VDD3P3_RTC		IE
19	GPIO14	VDD3P3_RTC		IE
20	VDD3P3_RTC			
21	XTAL_32K_P	VDD3P3_RTC		
22	XTAL_32K_N	VDD3P3_RTC		
23	GPIO17	VDD3P3_RTC		IE
24	GPIO18	VDD3P3_RTC		IE
25	GPIO19	VDD3P3_RTC		
26	GPIO20	VDD3P3_RTC	USB_PU	USB_PU

No.	Name	Power	At Reset	After Reset
27	GPIO21	VDD3P3_RTC		
28	SPICS1	VDD_SPI	IE, WPU	IE, WPU
29	VDD_SPI			
30	SPIHD	VDD_SPI	IE, WPU	IE, WPU
31	SPIWP	VDD_SPI	IE, WPU	IE, WPU
32	SPICS0	VDD_SPI	IE, WPU	IE, WPU
33	SPICLK	VDD_SPI	IE, WPU	IE, WPU
34	SPIQ	VDD_SPI	IE, WPU	IE, WPU
35	SPID	VDD_SPI	IE, WPU	IE, WPU
36	SPICLK_N	VDD_SPI / VDD3P3_CPU	IE	IE
37	SPICLK_P	VDD_SPI / VDD3P3_CPU	IE	IE
38	GPIO33	VDD_SPI / VDD3P3_CPU		IE
39	GPIO34	VDD_SPI / VDD3P3_CPU		IE
40	GPIO35	VDD_SPI / VDD3P3_CPU		IE
41	GPIO36	VDD_SPI / VDD3P3_CPU		IE
42	GPIO37	VDD_SPI / VDD3P3_CPU		IE
43	GPIO38	VDD3P3_CPU		IE
44	MTCK	VDD3P3_CPU		IE
45	MTDO	VDD3P3_CPU		IE
46	VDD3P3_CPU			
47	MTDI	VDD3P3_CPU		IE
48	MTMS	VDD3P3_CPU		IE
49	U0TXD	VDD3P3_CPU	IE, WPU	IE, WPU
50	U0RXD	VDD3P3_CPU	IE, WPU	IE, WPU
51	GPIO45	VDD3P3_CPU	IE, WPD	IE, WPD
52	GPIO46	VDD3P3_CPU	IE, WPD	IE, WPD
53	XTAL_N			
54	XTAL_P			
55	VDDA			
56	VDDA			
57	GND			

IE: input enabled. WPU: internal weak pull-up enabled. WPD: internal weak pull-down enabled. USB_PU: USB pull-up resistor enabled.
Source: [\[2\]](#), section GPIO.

3.2 Power-Up Glitches on Pins

Pin	Glitch ^a	Typical Time (μs)
GPIO1	Low-level glitch	60
GPIO2	Low-level glitch	60
GPIO3	Low-level glitch	60
GPIO4	Low-level glitch	60
GPIO5	Low-level glitch	60
GPIO6	Low-level glitch	60
GPIO7	Low-level glitch	60
GPIO8	Low-level glitch	60
GPIO9	Low-level glitch	60
GPIO10	Low-level glitch	60
GPIO11	Low-level glitch	60
GPIO12	Low-level glitch	60
GPIO13	Low-level glitch	60
GPIO14	Low-level glitch	60
XTAL_32K_P	Low-level glitch	60
XTAL_32K_N	Low-level glitch	60
GPIO17	Low-level glitch	60
GPIO18	Low-level / High-level glitch	60
GPIO19	Low-level glitch / High-level glitch ^b	60
GPIO20	Pull-down glitch / High-level glitch ^b	60

^a Low-level: pin outputs low for the period. High-level: pin outputs high for the period. Pull-down: pin is held by the internal weak pull-down for the period. ^b GPIO19 and GPIO20 each show two high-level glitches of about 60 μs during power-up; total duration and delay are 3.2 ms (GPIO19) and 2 ms (GPIO20).

Source: [2], section GPIO, table Power-Up Glitches on Pins.

3.3 GPIO Assignment

Current connection of every GPIO-capable pin, read from the schematic sheets. X marks a recommended change; the comment ID points to the table that follows. Reset state is at reset / after reset, from Section 3.1 (Hi-Z: no input, no pull). Sheets: MCU-3, Audio-1, Power-2, Periph-4.

Pin	Name	Net / connection	Sheet	Dir	Reset	External pull	Flags	Sw.	Cmt.
5	GPIO0	BOOT0: BOOT switch to GND via 1 kΩ	MCU-3	In	IE, WPU	10 kΩ pu	Strap (boot)		C1
6	GPIO1	GPIO1_DAC	Audio-1	I/O	IE	10 kΩ pd	Glitch		C2
7	GPIO2	GPIO2_DAC	Audio-1	I/O	IE	10 kΩ pd	Glitch		C2
8	GPIO3	TRI_STATE_24	Audio-1	Out	IE	10 kΩ pd (R8)	Strap (JTAG); glitch		C3
9	GPIO4	SDA_DAC / SDA_BQ	Audio-1, Power-2	I/O	Hi-Z	10 kΩ pu × 2	Glitch		C4
10	GPIO5	SCL_DAC / SCL_BQ	Audio-1, Power-2	Out	Hi-Z	10 kΩ pu × 2	Glitch		C4
11	GPIO6	DIM_LCD (2N7002 gate)	Periph-4	Out	Hi-Z	100 kΩ pd (gate)	Glitch		C19
12	GPIO7	DATA_CLK_DAC (I2S bit clock)	Audio-1	In	Hi-Z	None	Glitch		

Pin	Name	Net / connection	Sheet	Dir	Reset	External pull	Flags	Sw.	Cmt.
13	GPIO8	DATA1_DAC (I2S frame clock)	Audio-1	In	Hi-Z	None	Glitch		
14	GPIO9	DATA2_DAC (I2S data)	Audio-1	Out	Hi-Z / IE	None	Glitch		
15	GPIO10	PG_+3V3	Power-2	In	Hi-Z / IE	10 kΩ pu (R224)	Glitch		
16	GPIO11	PG_-3V3	Power-2	In	Hi-Z / IE	100 kΩ pu (R221)	Glitch		C16
17	GPIO12	SELECT (encoder push switch)	Periph-4	In	Hi-Z / IE	10 kΩ pd	Glitch; RTC		C12, C17
18	GPIO13	TRI_STATE_22	Audio-1	Out	Hi-Z / IE	10 kΩ pd (R7)	Glitch		
19	GPIO14	VOLUME+	Periph-4	In	Hi-Z / IE	10 kΩ pd	Glitch; RTC		C12, C17
21	GPIO15 (XTAL_32K_P)	ENC_B	Periph-4	In	Hi-Z	10 kΩ pu, 10 kΩ series, 10 nF	Glitch; RTC		C5
22	GPIO16 (XTAL_32K_N)	ENC_A	Periph-4	In	Hi-Z	10 kΩ pu, 10 kΩ series, 10 nF	Glitch; RTC		C5
23	GPIO17	VOLUME-	Periph-4	In	Hi-Z / IE	10 kΩ pd	Glitch; RTC		C12, C17
24	GPIO18	RESET_LCD	Periph-4	Out	Hi-Z / IE	None	Glitch (low / high)		C8
25	GPIO19	DN (USB D-)	Power-2	I/O	Hi-Z	22 Ω series	Glitch; USB		
26	GPIO20	DP (USB D+)	Power-2	I/O	USB_PU	22 Ω series	Glitch; USB		
27	GPIO21	POWER	Periph-4	In	Hi-Z	10 kΩ pd	RTC (wake)		C12
28	SPICS1 (GPIO26)	CD_SD (card detect)	Periph-4	In	IE, WPU	10 kΩ pu			C6
30	SPIHD (GPIO27)	SPI_HD; flash pin 7 (HOLD# / IO3)	MCU-3	I/O	IE, WPU		Flash		C9
31	SPIWP (GPIO28)	SPI_WP; flash pin 3 (WP# / IO2)	MCU-3	I/O	IE, WPU		Flash		C9
32	SPICS0 (GPIO29)	SPI_CS0; flash pin 1 (CS#)	MCU-3	Out	IE, WPU		Flash		C9
33	SPICLK (GPIO30)	SPICLK; 22 Ω series to flash pin 6 (SCLK)	MCU-3	Out	IE, WPU		Flash		C9
34	SPIQ (GPIO31)	SPI_Q; flash pin 2 (SO / IO1)	MCU-3	I/O	IE, WPU		Flash		C9
35	SPID (GPIO32)	SPI_D; flash pin 5 (SI / IO0)	MCU-3	I/O	IE, WPU		Flash		C9
36	SPICLK_N (GPIO48)	EN_LED (BD1604MUV EN)	Periph-4	Out	IE	Internal pd in driver	VDD_SPI domain		C7
37	SPICLK_P (GPIO47)	CLK_SD; 22 Ω series	Periph-4	Out	IE	None	VDD_SPI domain		C6
38	GPIO33	PLUG_DET	Audio-1	In	Hi-Z / IE	100 kΩ pu (R19), 10 kΩ series (R20)			
39	GPIO34	EN_SPEAKER	Audio-1	Out	Hi-Z / IE	50 kΩ pd (R31) via 54.9 kΩ (R29)			
40	GPIO35	SHDN_BB_3.9	Power-2	Out	Hi-Z / IE	1 MΩ pd (MCU-3)			C10
41	GPIO36	RESETB_DAC	Audio-1	Out	Hi-Z / IE	10 kΩ pd (R12)			C2
42	GPIO37	INT_BQ	Power-2	In	Hi-Z / IE	10 kΩ pu			
43	GPIO38	D/CX_LCD	Periph-4	Out	Hi-Z / IE	None			
44	MTCK (GPIO39)	SCL_LCD; 22 Ω series	Periph-4	Out	Hi-Z / IE	None	JTAG pin		
45	MTDO (GPIO40)	MOSI_LCD	Periph-4	Out	Hi-Z / IE	None	JTAG pin		
47	MTDI (GPIO41)	CMD_SD	Periph-4	I/O	Hi-Z / IE	10 kΩ pu	JTAG pin		C6
48	MTMS (GPIO42)	DAT0_SD	Periph-4	I/O	Hi-Z / IE	10 kΩ pu	JTAG pin		C6
49	U0TXD (GPIO43)	UART0 TX: debug console (test pad)	MCU-3	Out	IE, WPU	None	UART0 TX		C11
50	U0RXD (GPIO44)	UART0 RX: debug console (test pad)	MCU-3	In	IE, WPU	None	UART0 RX		C11
51	GPIO45	CS_LCD	Periph-4	Out	IE, WPD	10 kΩ pd	Strap (VDD_SPI)		C13
52	GPIO46	CE_BQ	Power-2	Out	IE, WPD	Internal WPD	Strap (boot)		C14

Source: [8]; reset states and flags from [2]; strapping from [3].

ID	Comment	Source
C1	Fitted: 10 kΩ pull-up on BOOT0, BOOT switch to GND through 1 kΩ (≈ 0.3 V when pressed), RESET switch from CHIP_PU to GND. Download mode: hold BOOT, press and release RESET, release BOOT.	[2], [3], [8], [9]
C2	10 kΩ pull-downs on GPIO1_DAC, GPIO2_DAC and RESETB_DAC confirmed: the DAC GPIO default mode shuts the DAC down when high.	[15]
C3	GPIO3 now carries TRI_STATE_24. Its strap selects the JTAG source only when the eFuse enabling it is burned; R8 gives a defined low at reset, and the power-up low glitch keeps the oscillator disabled.	[3], [8]
C4	Two 10 kΩ pull-up pairs on the I2C bus (DAC sheet R10 / R11, charger sheet): 5 kΩ effective. Acceptable at 100 kHz; one pair is redundant.	[8]
C5	Encoder (TT Electronics EN12-VS11AF18, right angle, push switch): ENC_A on GPIO16, ENC_B on GPIO15, both RTC-capable (wake). Per line: 10 kΩ pull-up, 10 kΩ series, 10 nF to GND on the GPIO side; ESD array on ENC_A, ENC_B and SELECT.	[8], [9], [20]
C6	microSD (Same Sky MSD-1-A), 1-bit SDMMC: CLK_SD on GPIO47 (22 Ω series at the MCU), CMD_SD on GPIO41, DAT0_SD on GPIO42, CD_SD on GPIO26 (10 kΩ pull-up; low with a card inserted). 10 kΩ pull-ups on CMD and DAT0 to DAT3; TVS array on CLK, CMD, DAT0 and DAT3. 4-bit not used (would need GPIO1, GPIO2 and ADDR_DAC).	[8], [9], [19]
C7	GPIO48 drives EN_LED (backlight on / off). The BD1604MUV EN input has an internal pull-down (≈ 18 μA at 5.5 V), so the backlight stays off while GPIO48 floats at reset. GPIO47 / GPIO48 sit in the VDD_SPI domain and are 3.3 V only while GPIO45 = 0 and the flash is 3.3 V.	[2], [9], [21]
C8	Power-up glitch pulses the LCD reset; harmless because firmware resets the panel at start-up.	[2]
C9	Flash connected per the GD25Q64E 8-WSON pinout; VCC on VDD_SPI with 0.1 μF and 1 μF; pins 4 and 9 to GND. SPIWP / SPIHD / SPICSO internal pull-ups hold WP#, HOLD# and CS# inactive at reset. SPICLK: 22 Ω series fitted (place at pin 33); an unpopulated shunt-C footprint is optional per the SPI EMC note.	[2], [8], [11], [13]
C10	1 MΩ pull-down (#1-123) fitted on the GPIO side of R203, per the LTC3440 datasheet; holds the analog rails off until firmware enables them.	[10], [8], [12]
C11	UP / DOWN buttons removed (the encoder covers up / down). GPIO43 / GPIO44 return to UART0: bring them to test pads (with GND) for a debug console, the only log output while TinyUSB owns USB. If left unconnected, no termination is needed.	[2], [8]
C12	Buttons (SELECT, POWER, VOLUME+, VOLUME-) are active high (10 kΩ pull-down, switch to +3V3_DIG). GPIO inputs are high impedance, so either polarity works. No hardware debounce (FW6).	[8], [9]
C13	Keep the pull-down. A pull-up (usual for CS) makes GPIO45 = 1 at reset and selects 1.8 V on VDD_SPI.	[2]
C14	CE low at reset keeps charging enabled. GPIO46 = 1 with GPIO0 = 0 is an invalid strap combination.	[3]
C15	All GPIOs assigned.	[8]
C16	R221 changed to 100 kΩ: 33 μA, within the LT3093 PG low specification (100 μA). PG reads low in shutdown (analog rails off), so R221 draws 33 μA continuously in standby.	[17], [8]
C17	SELECT (GPIO12), VOLUME+ (GPIO14) and VOLUME- (GPIO17) output a 60 μs low glitch at power-up. A button held during power-up shorts that pin to +3V3 through the switch for the glitch period. Low risk; 1 kΩ in series, as on DOWN, removes it.	[2], [8]
C18	Fixed: the SELECT node no longer connects to the +3V3_DIG button bus.	[8]

ID	Comment	Source
C19	GPIO6 drives the 2N7002 gate (DIM_LCD): high = bright, low or floating = dim. The 100 kΩ gate pull-down holds dim at reset, and the power-up low glitch only means dim while EN_LED is off. ADDR_DAC no longer reaches the MCU; the DAC ADDR pin is pulled up on DAP-Audio-1 (R9, 10 kΩ; author): chip address 0x92 write, 7-bit 0x49. Remove the ADDR_DAC port on that sheet.	[2], [8], [9], [22]

4. External Components

Ref	AL-PN	Part	Notes	Qty
U		Espressif ESP32-S3	56-QFN, 7 × 7 mm	1
Y	#1-118	ECS Inc. ECS-400-10-37B2-CKY-TR	40 MHz crystal, C _L 10 pF, ±10 ppm tolerance, ±10 ppm stability, ESR 40 Ω, 4-SMD. DigiKey 50-ECS-400-10-37B2-CKY-TRCT-ND. [4], [5]	1
C (XTAL)	#1-122	Murata GJM1555C1H120FB01D	12 pF, ±1 %, 50 V, COG, 0402. Crystal load capacitors, one per crystal pin (C1 / C4 in the guideline figure). DigiKey 490-11190-1-ND. [6]	2
L (XTAL_P)	#1-120	TDK MHQ1005075HA24N HTD25	24 nH, ±3 %, 250 mA, 700 mΩ max, 0402. Series component on XTAL_P; 24 nH is the guideline starting value. DigiKey 445-MHQ1005075HA24NHTD25CT-ND. [2], [7]	1
L (VDD3P3)	#1-121	TDK MHQ1005075HA2N0 BTD25	2 nH, 1 A, 50 mΩ max, 0402. LC filter inductor on the VDD3P3 rail (pins 2 and 3); guideline asks for ≥ 500 mA rating. DigiKey 445-MHQ1005075HA2N0BTD25CT-ND. [2], [7]	1
U (flash)	#1-124	GigaDevice GD25B64EWIGR	64 Mbit (8 MB) SPI NOR flash, quad I/O, 133 MHz, 2.7 to 3.6 V, -40 to 85 °C, 8-WSON (6 × 5 mm), exposed pad. Powered from VDD_SPI. Footprint SON127P600X500X80-9N. DigiKey 1970-GD25B64EWIGRCT-ND. [11], [13]	1
R (GPIO35)	#1-123	YAGEO RE0402BRE071ML	1 MΩ, ±0.1 %, 1/16 W, 0402, thick film. SHDN_BB_3.9 pull-down on the GPIO side of R203. DigiKey 13-RE0402BRE071MLCT-ND. [10], [12]	1
SW (BOOT, RESET)		C&K; PTS810 SJM 250 SMTR LFS	Top-actuated SMD tactile, 4.2 × 3.2 mm, same part as the front switches. BOOT on GPIO0, RESET on CHIP_PU; internal, used with the enclosure open. [9]	2

5. Equations and Results

Crystal load capacitors (C1 = C4)

$$C_L = \frac{C_1 C_4}{C_1 + C_4} + C_{\text{stray}} \quad (1)$$

$$C_1 = C_4 \Rightarrow C_1 = C_4 = 2(C_L - C_{\text{stray}}) \quad (2)$$

$$C_{L, \text{eff}} = \frac{C_1}{2} + C_{\text{stray}} \quad (3)$$

Source: equation (1) and C1 / C4 naming per [2], section External Crystal Clock Source. Equations (2) and (3) follow from (1) with C1 = C4.

Parameter	Value	Source
C _L (crystal load capacitance)	10 pF	[4], [5]
C _{stray} (PCB stray capacitance)	3 to 5 pF (assumed; the guideline gives no value)	Estimate
C1 = C4 required	10 to 14 pF	Eq. (2)
C1 = C4 selected	12 pF (#1-122), centre of the required range	Eq. (2)
C _{L,eff} with 12 pF	9 to 11 pF (C _L = 10 pF ± 1 pF)	Eq. (3)
Tuning	Set after assembly: TX tone test, measure 2.4 GHz frequency offset, trim to within ±10 ppm. Positive offset: increase load capacitance. Negative offset: decrease it.	[2]

Flash package and land pattern (8-WSON, 6 × 5 mm)

Symbol	Meaning	Min	Nom	Max
D	Body length (lead to lead direction)	5.90	6.00	6.10
E	Body width (pitch direction)	4.90	5.00	5.10
A	Height	0.70	0.75	0.80
e	Pitch		1.27	
b	Lead width	0.35	0.40	0.50
L	Lead length	0.50	0.60	0.75
D2	Exposed pad along D	3.30	3.40	3.50
E2	Exposed pad along E	3.90	4.00	4.10

All dimensions mm. Source: [13], Section 10.6 (GD25Q64E; same GigaDevice WSON8 6 × 5 mm outline, not yet confirmed in a GD25B64E datasheet). Exposed pad is floating.

Land pattern item	Value	Source
IPC-7351 name	SON127P600X500X80-9N (library #1-124)	Author
Pin order	Counter-clockwise from pin 1 viewed from the top: 1 to 4 down one side, 5 to 8 up the other; pin 8 opposite pin 1	[13]
Pinout	1 CS#, 2 SO / IO1, 3 WP# / IO2, 4 VSS, 5 SI / IO0, 6 SCLK, 7 HOLD# / IO3, 8 VCC, 9 exposed pad	[13]
Lead lands (reviewed)	≈ 1.01 × 0.39 mm, spanning 2.49 to 3.50 mm from centre	Author footprint
Exposed pad land (reviewed)	≈ 3.50 × 3.97 mm; ≈ 0.74 mm to the lead lands	Author footprint
Exposed pad paste	Split into windowpanes, ≈ 50 % coverage	Recommendation

6. Guideline Notes

Topic	Note	Source
SPI (EMC)	Place a series resistor or ferrite bead plus a capacitor to ground on the SPI_CLK trace. Where space allows, add the same series R and shunt C on the other SPI traces. Keep the RC / LC parts close to the chip pins.	[2], SPI
XTAL_P	Series component on XTAL_P, 24 nH initial value, adjusted after an overall test, to limit crystal harmonics reaching the RF path.	[2], Clock Source
VDD3P3	LC filter on the VDD3P3 rail to suppress high-frequency harmonics; inductor rated 500 mA or more. 10 µF on the rail recommended because TX bursts can collapse it.	[2], Analog Power Supply
Crystal	Crystal amplitude recommended above 500 mV. Frequency deviation beyond ±10 ppm degrades RF performance.	[2], Clock Source
CHIP_PU	RC delay recommended, typically R = 10 kΩ and C = 1 µF; must not float.	[2], Power-up and Reset
Strapping	GPIO0, GPIO3, GPIO45 and GPIO46 are strapping pins. GPIO0 and GPIO46 set boot mode; GPIO45 sets VDD_SPI voltage (0: 3.3 V, 1: 1.8 V) with default eFuses. GPIO46 = 1 with GPIO0 = 0 is an invalid combination.	[2], [3]

7. Firmware Requirements

Behaviour the hardware relies on. Each item is required unless marked optional.

ID	Requirement	Reason	Source
FW1	Power sequencing: drive SHDN_BB_3.9 high, wait for PG_+3V3 and PG_-3V3, then assert TRI_STATE_22 / 24. On shutdown, release TRI_STATE before dropping SHDN_BB_3.9.	Oscillators run on +3V3_ANALOG; a high on TRI_STATE (+3V3_DIG GPIO) with the rail off back-powers them through the enable input.	[8], [15]
FW2	Enable at most one oscillator (TRI_STATE_22 or TRI_STATE_24) at any time.	Both drive the DAC XI input.	[15]
FW3	Treat PG_+3V3 as valid only while the 3.9 V rail is enabled.	The LT3042 PG is high impedance in shutdown and reads good while the rail is off.	[16], [18]
FW4	Drive EN_SPEAKER high only $\approx$ 214 ms after enabling the analog rails; drive it low on a PG fault.	Rails must be settled before the OPA1622 output is enabled; DC-fault protection.	[16], [18]
FW5	UART0 console on GPIO43 / GPIO44 for logging (115200 baud default).	USB-Serial-JTAG is unavailable once TinyUSB MSC owns the USB PHY.	[2], [8]
FW6	Debounce all buttons in software.	No hardware debounce fitted.	[8]
FW7	Run I2C bus recovery (clock out SCL until SDA is released, then STOP) at start-up.	GPIO4 / GPIO5 power-up glitches can present a false START to the DAC and charger.	[2]
FW8	Keep GPIO1_DAC / GPIO2_DAC low (input, or driven low) until DAC Register 8 reconfigures them.	Default DAC mode shuts the DAC down when either pin is high.	[15]
FW9	Address the DAC at 7-bit I2C address 0x49 (0x92 write / 0x93 read).	ADDR pulled up by R9; no longer driven by the MCU (C19).	[22], [8]
FW10	Run the SD card in 1-bit SDMMC mode at up to 40 MHz; treat CD_SD low as card present, debounce it, and mount or unmount on change.	Hardware is 1-bit; CD_SD has a 10 k Ω pull-up and the switch pulls it low with a card inserted.	[8], [19]
FW11	Decode ENC_A / ENC_B with the PCNT peripheral and enable its glitch filter; one detent = one full quadrature cycle (4 edges; 12 detents, 12 pulses per turn).	Contact bounce up to 5 ms remains after the 10 nF RC filter.	[20]
FW12	Backlight: EN_LED high only while the display is on; DIM_LCD high for bright (19.6 mA per LED), low for dim (4.8 mA per LED). Do not PWM EN_LED.	EN PWM needs a 100 to 200 Hz period at low duty, which puts 80 mA steps in the audio band on SYS and ground.	[9], [21]
FW13	Use the BQ24250 INT (GPIO37) and its status register as VBUS-present for TinyUSB: enable the USB pull-up only with VBUS present and stop the MSC session on removal.	A self-powered USB device must sense VBUS; no GPIO is free for a VBUS divider.	[8]
FW14	Optional: lower GPIO drive strength on SPICLK / SCL_LCD if EMI or audio coupling appears.	Slower edges with no added parts.	Advice

8. Open Problems

#	Problem
1	Optional: 1 k Ω series on SELECT, VOLUME+ and VOLUME- against the power-up glitch (C17).
2	Schematic housekeeping: flash value text reads 64Bit.

Source: [2], [8], [9], [17].

9. Revision Log

Rev	Date	Change
0	2026-09-29	Initial issue: ESP32-S3 base chip, 40 MHz crystal, design references, schematic review items.
1	2026-09-29	Crystal changed to ECS-400-10-37B2-CKY-TR (10 pF). Load capacitors, XTAL_P series inductor and VDD3P3 filter inductor added. Pin default configuration and power-up glitch tables added. Guideline notes added (SPI EMC). Guideline source moved to the online ESP32-S3 Hardware Design Guidelines. Open problems 1, 8, 9 updated; 11 and 12 added. All entries cited.
2	2026-09-29	Crystal load capacitors changed to 12 pF GJM1555C1H120FB01D (#1-122); $C_{L,eff}$ 9 to 11 pF recorded. Open problem 1 (load capacitor value) closed; open problems renumbered.
3	2026-09-29	GPIO assignment table (Section 3.3) from the current schematic sheets, with recommended changes and comments. Closed: LCD off flash pins, CHIP_PU RC, VDDA supply, I2C crossing, strapping (EN1_BQ removed, CE_BQ on GPIO46, CS_LCD pull-down on GPIO45), power-up glitch review. Added: boot access, SHDN_BB_3.9 pull-down, UOTXD series resistor, GPIO budget, button polarity, flash-pin stubs.
4	2026-09-29	Flash selected: GigaDevice GD25B64EWIGR (8 MB, quad SPI, WSON-8). SHDN_BB_3.9 pull-down part #1-123 (1 M Ω). BOOT and RESET buttons: C&K; PTS810 (reused). Flash pin mapping in the GPIO table. Open problems updated.
5	2026-09-30	Flash added to library (#1-124, SON127P600X500X80-9N). Package dimensions and pinout from the GD25Q64E datasheet recorded; land pattern review results recorded. Open problem 1 narrowed.
6	2026-09-30	Reviewed against DAP-MCU-3 of 2026-09-30. Closed: flash datasheet and footprint, PSRAM (not used), SHDN_BB_3.9 pull-down, flash wiring, supply decoupling. GPIO table updated (flash nets, GPIO35 pull-down, GPIO0 BOOT wiring, GPIO26 free). Added: BOOT switch rail, SPICLK EMC footprint, housekeeping; C15.
7	2026-10-01	Firmware Requirements section added (FW1 to FW10). Closed: boot access, DOWN series resistor, SPICLK series resistor, button polarity (active high kept). Added: PG_-3V3 pull-up (C16), button power-up glitch (C17), Peripherals polarity update.
8	2026-10-01	SD and encoder assigned: CLK_SD GPIO47, CMD_SD GPIO41, DAT0_SD GPIO42, CD_SD GPIO26, ENC_A GPIO16, ENC_B GPIO15; SELECT moved to GPIO12 (encoder switch), TRI_STATE_24 to GPIO3. Storage requirement set to 1-bit SDMMC. FW10 and FW11 added; optional drive-strength item renumbered FW12. Closed: GPIO budget, microSD selection, Peripherals polarity, C2. Added: SELECT tied to +3V3_DIG (C18).
9	2026-10-02	Backlight pins assigned: EN_LED on GPIO48, DIM_LCD on GPIO6 (ADDR_DAC moved to a hardware strap). FW9 rewritten for the strapped address; FW12 (backlight control) added, optional drive-strength item now FW13. Closed: SELECT tied to +3V3_DIG, backlight pin. Added: DAC address strap check.
10	2026-10-02	DAC ADDR tied to GND: I2C address 0x48 (7-bit) recorded in C19 and FW9. Closed: DAC address strap.

Rev	Date	Change
11	2026-10-02	DAC ADDR reverted to the R9 pull-up: I2C address 0x49 (7-bit), matching the DAC Clocking document (0x92 write).
12	2026-10-03	R221 100 kΩ (closes C16). FW13 added (VBUS-present from the charger); optional drive-strength item now FW14. Designators annotated on the schematic. Closed: PG_-3V3 pull-up, annotation.
13	2026-10-04	UP / DOWN buttons removed; GPIO43 / GPIO44 return to UART0 (debug console, C11, FW5). Encoder changed to TT Electronics EN12-VS11AF18 (12 detents, 12 pulses; FW11).

10. Sources

1. [ESP32-S3 Hardware Design Guidelines \(Espressif\)](#)
2. [ESP32-S3 Hardware Design Guidelines, Schematic Checklist \(Espressif\)](#)
3. [ESP32-S3 Series Datasheet \(Espressif\)](#)
4. [ECS-400-10-37B2-CKY-TR Product Page \(DigiKey\)](#)
5. [ECS-400-10-37B2-CKY-TR Product Page \(ECS Inc.\)](#)
6. [GJM1555C1H120FB01D Product Page \(DigiKey\)](#)
7. [MHQ1005075HA Series Inductors \(TDK, via DigiKey\)](#)
8. DAP schematic sheets DAP-MCU-3 and DAP-Peripherals-4 (annotated), DAP-Audio-1, DAP-Power-2 (author, 2026-10-03)
9. Peripherals: Working Document, Rev 13
10. LTC3440 datasheet, SHDN/SS pull-down value (as reported by the author)
11. [GD25B64EWIGR Product Listing \(Dasenic\)](#)
12. RE0402BRE071ML, author library entry #1-123 (DigiKey 13-RE0402BRE071MLCT-ND)
13. [GD25Q64E Datasheet \(GigaDevice\), Sections 2 and 10.6](#)
14. [GD25B64C Product Page \(GigaDevice\)](#)
15. DAC Clocking and I2S Interface (ES9038Q2M): Working Document, Rev 10
16. Positive LDO (LT3042): Working Document, Rev 7
17. Negative LDO (LT3093): Working Document, Rev 10
18. Output Stage (OPA1622): Working Document, Rev 5
19. [MSD-1-A Product Page \(Same Sky\)](#)
20. [EN12-VS11AF18 Product Page \(DigiKey\)](#)
21. [BD1604MUV / BD1604MVV Datasheet \(ROHM, via Mouser\)](#)
22. ES9038Q2M Datasheet v1.4 (ESS Technology), p. 14 I2C address; project file